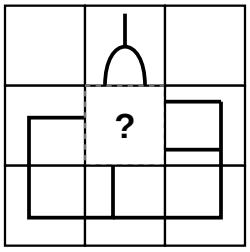
Math Kangaroo Practice - Grades 3-4

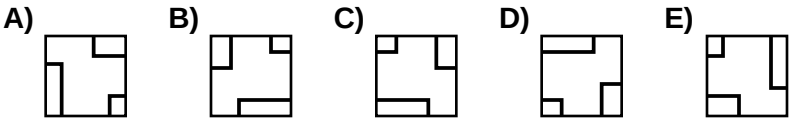
Practice set for 2002 #1
Five practice problems per selected problem.

Set 1 - Missing puzzle tile

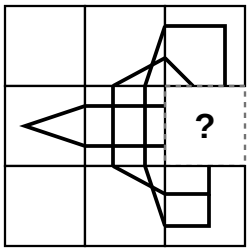
Based on 2002, Problem 1

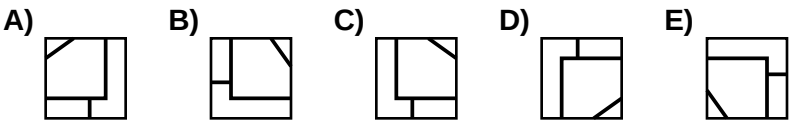
Practice 1.1 A garden path and archway are drawn across the 3 × 3 grid below, with the middle tile missing. Which tile (A through E) completes the picture?



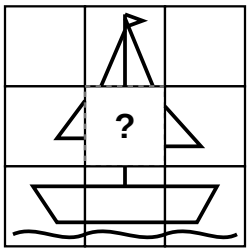


Practice 1.2 One square tile is missing from the geometric pattern below. Which tile (A through E) fits in the empty square marked with a question mark (?)?



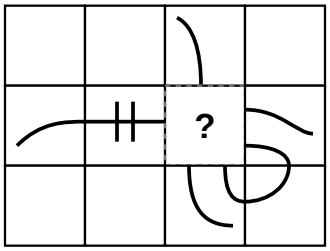


Practice 1.3 Which of the squares below is missing from the picture of the sailboat?



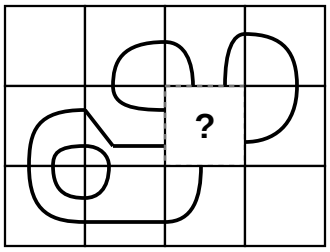
A)
B)
C)
D)
E)

Practice 1.4 Streams flow across the 3 × 4 map below, and one square tile is missing. Which tile (A through E) fits in the square marked with a question mark (?)?



A)
B)
C)
D)
E)

Practice 1.5 Park paths wind across the 3 × 4 map below, and one square tile has been removed. Which of the tiles (A through E) belongs in the square marked with a question mark (?)?



A)
B)
C)
D)
E)

Answer Key & Explanations

For generated/2026-10-19-20-18-41.pdf (Practice set for 2002 #1)

Solutions & kid-friendly explanations

1.1: B In the center square, two arch lines enter from the top (at 0.25 and 0.75), one line enters from the left (at 0.60), one enters from the bottom (at 0.35), and two exit through the right (at 0.20 and 0.80). Only tile B connects all of these endpoints cleanly.

1.2: A Looking at the neighbours of the missing square on the right: two vertical lines enter from the top (at 0.35 and 0.75), two horizontal lines enter from the left (at 0.25 and 0.75), one vertical line enters from the bottom (at 0.55), and nothing touches the outer right edge. Only tile A matches.

1.3: C In the center square, the vertical mast runs straight from the top center (0.50) to the bottom center (0.50), the left jib sail is higher than the right mainsail, and both horizontal booms meet the mast. Only square C matches all four edges of the missing center square.

1.4: C In the missing square, one stream enters from the left (at 0.55) and curves down to the bottom (at 0.30), one enters from the top (at 0.45) and curves to the upper-right (at 0.70), and a third connects the bottom-right (0.75) to the lower-right edge (0.25). Only tile C matches all six boundary crossings.

1.5: D Checking the four edges of the missing square: two paths enter from the top (at 0.35 and 0.75), two enter from the left (at 0.25 and 0.70), one exits through the bottom (at 0.45), and one exits through the right (at 0.30). Only tile D connects all six boundary points properly.